

# A statistical analysis of Congestion Control algorithms

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# Goals

- What is congestion control
- Challenges in measuring performance
- Measuring Congestion Control Algorithms
- Demo

# What is Congestion Control



# What is a Congestion control

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- Try to "best" utilize the link
- No visibility into the network
- Only 4 signals available:
  - throughput
  - loss
  - latency
  - jitter



# What is a Congestion control

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- Goal
  - send at the optimal rate
  - ensure fairness
  - minimize delay
  - prevent congestion collapse

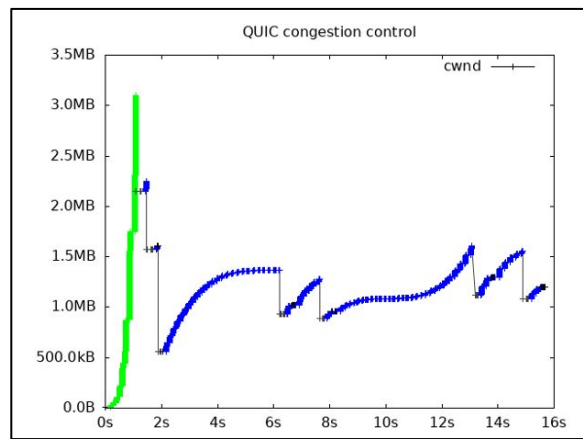
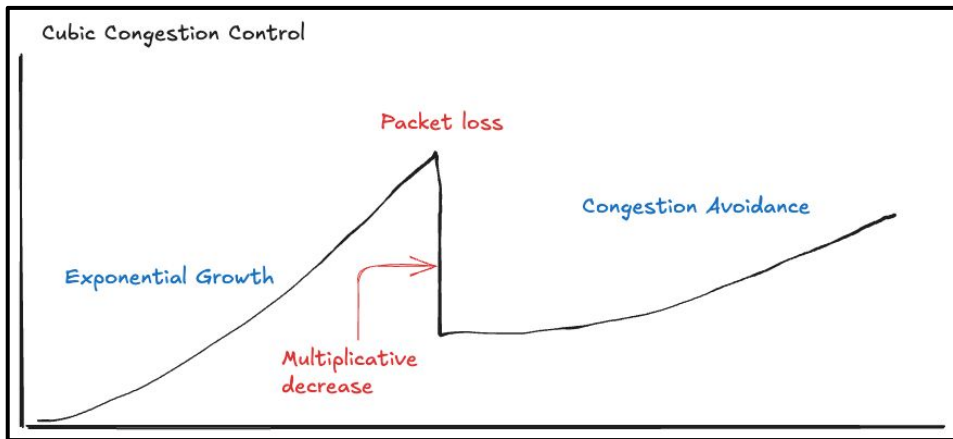
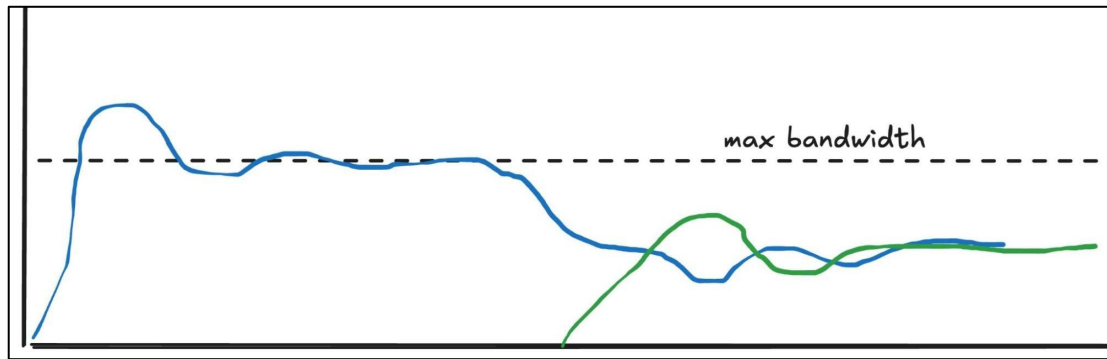
## Types of Congestion control algo.

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- Loss based (Cubic): looks at the loss signal
- Model based (BBR): model the network; reduce the standing queue
- Delay based (Copa): looks at the evolution of delay

# Building intuition: Cubic

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# Challenges in measuring performance





# Challenges: Network

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- Behavior depends on both server and client
- Other co-existing flows
- Wide range of use case
  - Loss:     DC (~.1%)           WLAN (0-1%, up to 2%)
  - RTT:      DC (<1ms)          WLAN (25ms - 300ms)
  - Jitter: DC (~1ms)           WLAN (20 - 50ms)

# Challenges: System and Algorithm

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- OS: overload, context switches, interrupts. Shared CPU, Mem etc.
- NIC buffering: overload, interrupts
- Correctness: partially broken can be hard to debug
- BBR:
  - Pacing: micro-sec granularity (milli-sec is ok)
  - Send quantum: GSO, GRO are tricky to get right

# Key insight

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- The network is not deterministic
- The process is not deterministic
- A test run is not deterministic
- **Q:** How do we find the signal in the noise?
  - Apply statistical analysis

# Measuring Congestion Control Algorithms

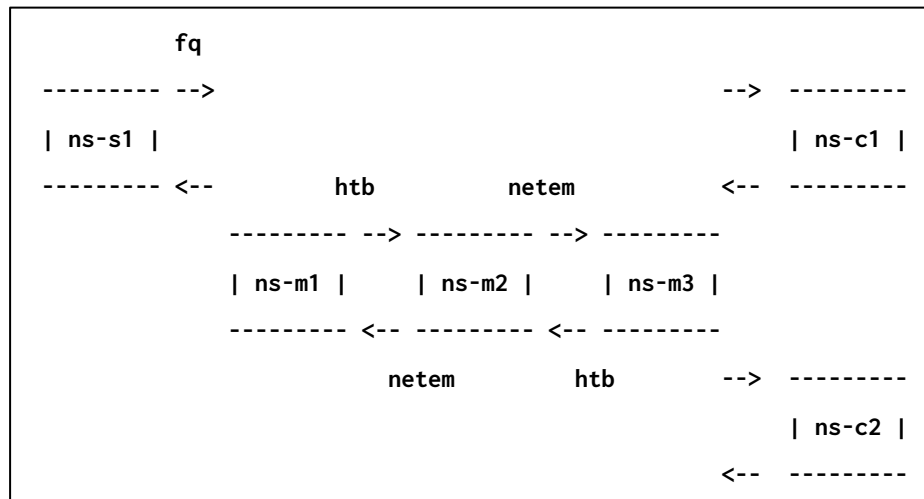


# The testing environment

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- netns: virtual network namespace
- veth: virtual Eth
- tc-htb: rate limit
- tc-netem: delay + loss
- tc-fq: pacing

code

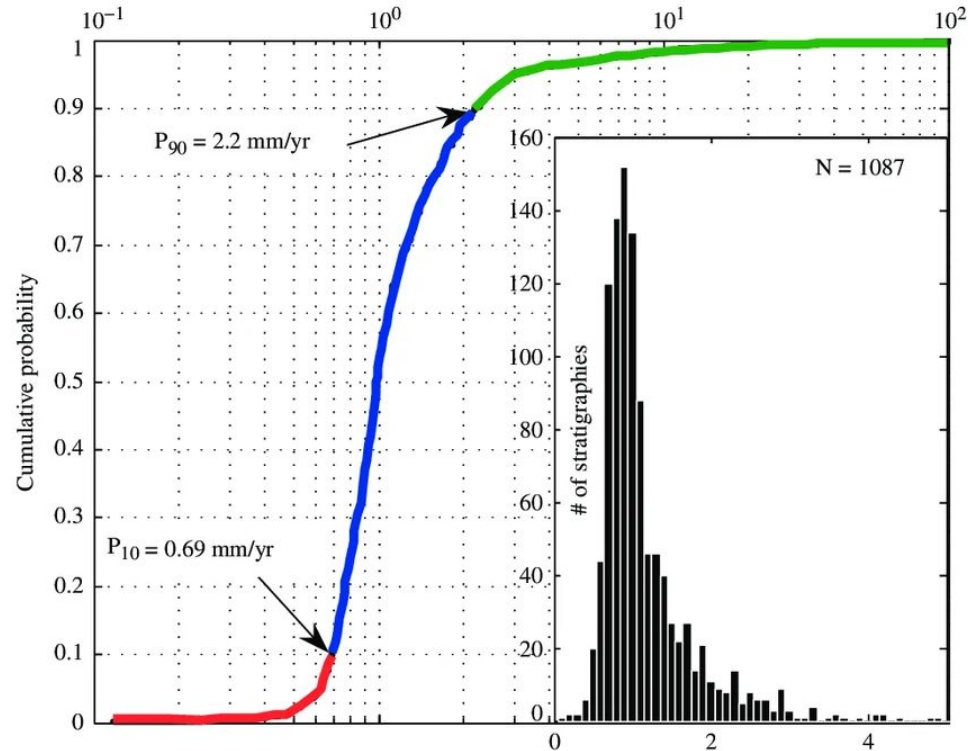




# Cumulative distribution function (CDF)

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- Visualize and easily compare probability distributions
- Tail analysis requires 100+ samples



# Presenting Boar



# Boar

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- App: [boar.toidiu.com](http://boar.toidiu.com)
  - sample data: [data](#)
- Code: [github.toidiu.com/boar](https://github.com/toidiu/boar)



Thank you



# Q&A

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